

Learning of state-space models with highly informative observations

A tempered Sequential Monte Carlo solution

Andreas Svensson, Thomas B. Schön, and Fredrik Lindsten

Presentation by: Finn van Vlaanderen

Seminar 'Sequential Monte Carlo Methods'

May 29, 2026

Highly informative observations

Kernels P_0^θ, P^θ on measurable space $\mathcal{X}$, parameter $\theta \in \Theta$,

$$x_0 \sim P_0^\theta(dx_0),$$

$$x_t|x_{t-1} \sim P^\theta(dx_t, x_{t-1}), \quad t = 1, \dots, T$$

$$y_t|x_t \sim N(g^\theta(x_t), \sigma^2), \quad t = 0, \dots, T$$

deterministic $g^\theta: \mathcal{X} \rightarrow \mathcal{X}$, σ^2 negligibly small relative to latent state noise.

Highly informative observations

Kernels P_0^θ, P^θ on measurable space $\mathcal{X}$, parameter $\theta \in \Theta$,

$$x_0 \sim P_0^\theta(dx_0),$$

$$x_t|x_{t-1} \sim P^\theta(dx_t, x_{t-1}), \quad t = 1, \dots, T$$

$$y_t|x_t \sim N(g^\theta(x_t), \sigma^2), \quad t = 0, \dots, T$$

deterministic $g^\theta: \mathcal{X} \rightarrow \mathcal{X}$, σ^2 negligibly small relative to latent state noise.

Example

When measurements can be made with good precision, but we lack knowledge of the (physical) system, or system is noisy

Highly informative observations

Kernels P_0^θ, P^θ on measurable space $\mathcal{X}$, parameter $\theta \in \Theta$,

$$x_0 \sim P_0^\theta(dx_0),$$

$$x_t|x_{t-1} \sim P^\theta(dx_t, x_{t-1}), \quad t = 1, \dots, T$$

$$y_t|x_t \sim N(g^\theta(x_t), \sigma^2), \quad t = 0, \dots, T$$

deterministic $g^\theta: \mathcal{X} \rightarrow \mathcal{X}$, σ^2 negligibly small relative to latent state noise.

Probabilistic learning

Fixed sample $y_{0:T} \in \mathcal{X}^{T+1}$ from state-space model, can we do inference on θ ?

► Bayesian approach: define prior $\nu(d\theta)$ on Θ , do inference on the marginal posterior

► Marginal posterior is given by

$$\mathbb{P}_T(d\theta|Y_{0:T} = y_{0:T}) = \frac{p_T^\theta(y_{0:T})\nu(d\theta)}{\int p_T^\theta(y_{0:T})\nu(d\theta)},$$

with $p_T^\theta(y_{0:T})$ the likelihood of $y_{0:T}$.

► Likelihoods $p_T^\theta(y_{0:T})$ typically intractable

Example

When measurements can be made with good precision, but we lack knowledge of the (physical) system, or system is noisy

Highly informative observations

Kernels P_0^θ, P^θ on measurable space $\mathcal{X}$, parameter $\theta \in \Theta$,

$$x_0 \sim P_0^\theta(dx_0),$$

$$x_t|x_{t-1} \sim P^\theta(dx_t, x_{t-1}), \quad t = 1, \dots, T$$

$$y_t|x_t \sim N(g^\theta(x_t), \sigma^2), \quad t = 0, \dots, T$$

deterministic $g^\theta: \mathcal{X} \rightarrow \mathcal{X}$, σ^2 negligibly small relative to latent state noise.

Example

When measurements can be made with good precision, but we lack knowledge of the (physical) system, or system is noisy

Probabilistic learning

Fixed sample $y_{0:T} \in \mathcal{X}^{T+1}$ from state-space model, can we do inference on θ ?

► Bayesian approach: define prior $\nu(d\theta)$ on Θ , do inference on the marginal posterior

► Marginal posterior is given by

$$\mathbb{P}_T(d\theta|Y_{0:T} = y_{0:T}) = \frac{p_T^\theta(y_{0:T})\nu(d\theta)}{\int p_T^\theta(y_{0:T})\nu(d\theta)},$$

with $p_T^\theta(y_{0:T})$ the likelihood of $y_{0:T}$.

► Likelihoods $p_T^\theta(y_{0:T})$ typically intractable

Takeaway

To perform parameter learning in state-space models, we have to be careful! First: How to even estimate likelihoods?

Idea

- ▶ Use some MCMC sampler to sample from the posterior.
- ▶ Likelihood $p_T^\theta(y_{1:T})$, needed in the sampler, is often unknown.

Idea

- ▶ Use some MCMC sampler to sample from the posterior.
- ▶ Likelihood $p_T^\theta(y_{1:T})$, needed in the sampler, is often unknown.

Algorithm: generic particle filter

Input: N particles, Feynman-Kac model

$\mathbb{M}_0^\theta, M_{1:T}^\theta, G_0^\theta, G_{1:T}^\theta$

Output: Particle trajectories $(X_{0:T}^{1:N}, A_{1:T}^{1:N})$

- 1: $X_0^n \sim \mathbb{M}_0^\theta(\mathrm{d}x_0)$
- 2: $w_0^n \leftarrow G_0^\theta(X_0^n)$
- 3: $W_0^n \leftarrow w_0^n / \sum_{m=0}^N w_0^m$
- 4: **for** $t = 1$ to T **do**
- 5: $A_t^{1:N} \sim \text{resample}(W_t^{1:N})$
- 6: $X_t^n \sim M_t^\theta(X_{t-1}^{A_t^n}, \mathrm{d}x_t)$
- 7: $w_t^n \leftarrow G_t(X_{t-1}^{A_t^n}, X_t^n)$
- 8: $W_t^n \leftarrow w_t^n / \sum_{m=1}^N w_t^m$
- 9: **end for**

Idea

- ▶ Use some MCMC sampler to sample from the posterior.
- ▶ Likelihood $p_T^\theta(y_{1:T})$, needed in the sampler, is often unknown.

Algorithm: generic particle filter

Input: N particles, Feynman-Kac model $\mathbb{M}_0^\theta, M_{1:T}^\theta, G_0^\theta, G_{1:T}^\theta$

Output: Particle trajectories $(X_{0:T}^{1:N}, A_{1:T}^{1:N})$

- 1: $X_0^n \sim \mathbb{M}_0^\theta(dx_0)$
- 2: $w_0^n \leftarrow G_0^\theta(X_0^n)$
- 3: $W_0^n \leftarrow w_0^n / \sum_{m=0}^N w_0^m$
- 4: **for** $t = 1$ **to** T **do**
- 5: $A_t^{1:N} \sim \text{resample}(W_t^{1:N})$
- 6: $X_t^n \sim M_t^\theta(X_{t-1}^{A_t^n}, dx_t)$
- 7: $w_t^n \leftarrow G_t(X_{t-1}^{A_t^n}, X_t^n)$
- 8: $W_t^n \leftarrow w_t^n / \sum_{m=1}^N w_t^m$
- 9: **end for**

Unbiased likelihood estimate

- ▶ From Chopin et al.: unbiased estimate of normalization constant L_T is

$$L_T^N(\theta, X_{0:T}^{1:N}, A_{1:T}^{1:N}) = \prod_{t=0}^T \frac{1}{N} \sum_{n=0}^N w_t^n$$

- ▶ Can use this if $L_T = p_T^\theta(y_{0:T})$ in Feynman-Kac model (ex. in bootstrap or guided).
- ▶ Notation: write $Z = (X_{0:T}^{1:N}, A_{1:T}^{1:N})$, $Z \sim K^\theta(dz)$, kernel on measurable space $\mathcal{Z}$.
- ▶ L_T^N is unbiased in the sense that

$$\mathbb{E}_{Z \sim K^\theta(\cdot)} L_T^N(\theta, Z) = L_T$$

Algorithm: bootstrap particle filter

Input: N particles, θ **Output:** Particle trajectories $(X_{0:T}^{1:N}, A_{1:T}^{1:N})$

- 1: $X_0^n \sim P_0^\theta(\mathrm{d}x_0)$
- 2: $w_0^n \leftarrow \mathrm{N}(y_0 | g^\theta(X_0^n), \sigma^2)$
- 3: $W_0^n \leftarrow w_0^n / \sum_{m=0}^N w_0^m$
- 4: **for** $t = 1$ **to** T **do**
- 5: $A_t^{1:N} \sim \text{resample}(W_t^{1:N})$
- 6: $X_t^n \sim P^\theta(X_{t-1}^{A_t^n}, \mathrm{d}x_t)$
- 7: $w_t^n \leftarrow \mathrm{N}(y_t | g^\theta(X_t^n), \sigma^2)$
- 8: $W_t^n \leftarrow w_t^n / \sum_{m=1}^N w_t^m$
- 9: **end for**

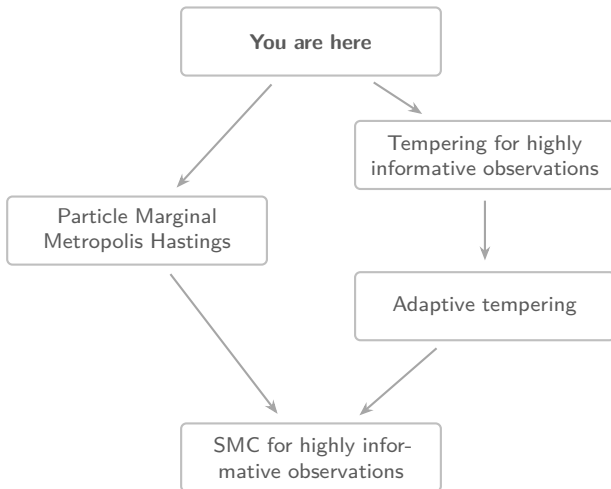
Algorithm: bootstrap particle filter

Input: N particles, θ **Output:** Particle trajectories $(X_{0:T}^{1:N}, A_{1:T}^{1:N})$

- 1: $X_0^n \sim P_0^\theta(dx_0)$
- 2: $w_0^n \leftarrow N(y_0 | g^\theta(X_0^n), \sigma^2)$
- 3: $W_0^n \leftarrow w_0^n / \sum_{m=0}^N w_0^m$
- 4: **for** $t = 1$ **to** T **do**
- 5: $A_t^{1:N} \sim \text{resample}(W_t^{1:N})$
- 6: $X_t^n \sim P^\theta(X_{t-1}^{A_t^n}, dx_t)$
- 7: $w_t^n \leftarrow N(y_t | g^\theta(X_t^n), \sigma^2)$
- 8: $W_t^n \leftarrow w_t^n / \sum_{m=1}^N w_t^m$
- 9: **end for**

The problem

- ▶ **Weight degeneracy:** when observation density is sharply peaked, mappings $X_t^n \mapsto N(y_t | g^\theta(X_t^n), \sigma^2)$ tightly concentrated around points where $g^\theta(X_t^n) \approx y_t$.
- ▶ Particles rarely ever hit observations
- ▶ When observation is found, sample collapses
- ▶ Likelihood estimates become highly variable (even if unbiased)
- ▶ Gets worse when parameter doesn't explain observations well



Idea

- ▶ Just replace likelihoods $p_T^\theta(y_{0:T})$ in MH with unbiased estimates $L_T^N(\theta, z)$.
- ▶ Will $\mathbb{P}_T(d\theta | Y_{0:T} = y_{0:T})$ still be the stationary distribution?

Idea

- ▶ Just replace likelihoods $p_T^\theta(y_{0:T})$ in MH with unbiased estimates $L_T^N(\theta, z)$.
- ▶ Will $\mathbb{P}_T(d\theta | Y_{0:T} = y_{0:T})$ still be the stationary distribution?

Algorithm: PMMH sampler (one step)

Input: N particles, θ ,
 PF trajectories $Z = (X_{0:T}^{1:N}, A_{0:T}^{1:N})$ for θ
Output: next parameter and PF trajectories

- 1: $\tilde{\theta} \sim Q(d\tilde{\theta}, \theta)$
- 2: $(\tilde{X}_{0:T}^{1:N}, \tilde{A}_{0:T}^{1:N}) := \tilde{Z} \sim K^{\tilde{\theta}}(dz)$
- 3: $U \sim \text{Unif}([0, 1])$
- 4: $\alpha \leftarrow \frac{\nu(\tilde{\theta}) L_T^N(\tilde{\theta}, \tilde{X}_{0:T}^{1:N}, \tilde{A}_{1:T}^{1:N}) q(\theta | \tilde{\theta})}{\nu(\theta) L_T^N(\theta, X_{0:T}^{1:N}, A_{1:T}^{1:N}) q(\tilde{\theta} | \theta)}$
- 5: **if** $U \leq \min(\alpha, 1)$ **then**
- 6: **return** $\tilde{\theta}, \tilde{X}_{0:T}^{1:N}, \tilde{A}_{1:T}^{1:N}$
- 7: **else**
- 8: **return** $\theta, X_{0:T}^{1:N}, A_{1:T}^{1:N}$

Idea

- ▶ Just replace likelihoods $p_T^\theta(y_{0:T})$ in MH with unbiased estimates $L_T^N(\theta, z)$.
- ▶ Will $\mathbb{P}_T(d\theta | Y_{0:T} = y_{0:T})$ still be the stationary distribution?

What's the stationary distribution?

- ▶ PMMH is just ordinary MH on an extended space $\Theta \times \mathcal{Z}$;
- ▶ with proposal

$$Q(d\tilde{\theta}, \theta) K^{\tilde{\theta}}(dz)$$

- ▶ and stationary distribution

$$\pi(d\theta, dz) = \frac{\nu(d\theta) L_T^N(\theta, z) K^\theta(dz)}{\int_{\Theta \times \mathcal{Z}} \nu(d\theta) L_T^N(\theta, z) K^\theta(dz)}$$

Algorithm: PMMH sampler (one step)

Input: N particles, θ ,
 PF trajectories $Z = (X_{0:T}^{1:N}, A_{0:T}^{1:N})$ for θ
Output: next parameter and PF trajectories

- 1: $\tilde{\theta} \sim Q(d\tilde{\theta}, \theta)$
- 2: $(\tilde{X}_{0:T}^{1:N}, \tilde{A}_{0:T}^{1:N}) := \tilde{Z} \sim K^{\tilde{\theta}}(dz)$
- 3: $U \sim \text{Unif}([0, 1])$
- 4: $\alpha \leftarrow \frac{\nu(\tilde{\theta}) L_T^N(\tilde{\theta}, \tilde{X}_{0:T}^{1:N}, \tilde{A}_{1:T}^{1:N}) q(\theta | \tilde{\theta})}{\nu(\theta) L_T^N(\theta, X_{0:T}^{1:N}, A_{1:T}^{1:N}) q(\tilde{\theta} | \theta)}$
- 5: **if** $U \leq \min(\alpha, 1)$ **then**
- 6: **return** $\tilde{\theta}, \tilde{X}_{0:T}^{1:N}, \tilde{A}_{1:T}^{1:N}$
- 7: **else**
- 8: **return** $\theta, X_{0:T}^{1:N}, A_{1:T}^{1:N}$

What's the stationary distribution?

▶ and stationary distribution

$$\pi(d\theta, dz) = \frac{\nu(d\theta) L_T^N(\theta, z) K^\theta(dz)}{\int_{\Theta \times \mathcal{Z}} \nu(d\theta) L_T^N(\theta, z) K^\theta(dz)}$$

▶ the posterior $\mathbb{P}_T(d\theta | Y_{0:T} = y_{0:T})$ is its marginal:

$$\begin{aligned} \int_{z \in \mathcal{Z}} \pi(d\theta, dz) &\propto \nu(d\theta) \int_{\mathcal{Z}} L_T^N(\theta, z) K^\theta(dz) \\ &= \nu(d\theta) \mathbb{E}_{z \sim K^\theta(dz)} L_T^N(\theta, z) \\ &= \nu(d\theta) p_T^\theta(y_{0:T}) \end{aligned}$$

because L_T^N is unbiased.

▶ Associated Markov kernel

$$M(d\tilde{\theta}, d\tilde{z}, \theta, z)$$

is an invariant kernel for $\pi(d\theta, dz)$.

Algorithm: PMMH sampler (one step)

Input: N particles, θ ,

PF trajectories $Z = (X_{0:T}^{1:N}, A_{0:T}^{1:N})$ for θ

Output: next parameter and PF trajectories

- 1: $\tilde{\theta} \sim Q(d\tilde{\theta}, \theta)$
- 2: $(\tilde{X}_{0:T}^{1:N}, \tilde{A}_{0:T}^{1:N}) := \tilde{Z} \sim K^{\tilde{\theta}}(dz)$
- 3: $U \sim \text{Unif}([0, 1])$
- 4: $\alpha \leftarrow \frac{\nu(\tilde{\theta}) L_T^N(\tilde{\theta}, \tilde{X}_{0:T}^{1:N}, \tilde{A}_{1:T}^{1:N}) q(\theta | \tilde{\theta})}{\nu(\theta) L_T^N(\theta, X_{0:T}^{1:N}, A_{1:T}^{1:N}) q(\tilde{\theta} | \theta)}$
- 5: **if** $U \leq \min(\alpha, 1)$ **then**
- 6: **return** $\tilde{\theta}, \tilde{X}_{0:T}^{1:N}, \tilde{A}_{1:T}^{1:N}$
- 7: **else**
- 8: **return** $\theta, X_{0:T}^{1:N}, A_{1:T}^{1:N}$

What's the stationary distribution?

▶ and stationary distribution

$$\pi(d\theta, dz) = \frac{\nu(d\theta) L_T^N(\theta, z) K^\theta(dz)}{\int_{\Theta \times \mathcal{Z}} \nu(d\theta) L_T^N(\theta, z) K^\theta(dz)}$$

▶ the posterior $\mathbb{P}_T(d\theta | Y_{0:T} = y_{0:T})$ is its marginal:

$$\begin{aligned} \int_{z \in \mathcal{Z}} \pi(d\theta, dz) &\propto \nu(d\theta) \int_{\mathcal{Z}} L_T^N(\theta, z) K^\theta(dz) \\ &= \nu(d\theta) \mathbb{E}_{z \sim K^\theta(dz)} L_T^N(\theta, z) \\ &= \nu(d\theta) p_T^\theta(y_{0:T}) \end{aligned}$$

because L_T^N is unbiased.

▶ Associated Markov kernel

$$M(d\tilde{\theta}, d\tilde{z}, \theta, z)$$

is an invariant kernel for $\pi(d\theta, dz)$.

Takeaway

- ▶ PMMH can sample from posteriors with intractable likelihoods.
- ▶ PMMH can be interpreted as a Metropolis-Hastings sampler on an extended parameter space $\Theta \times \mathcal{Z}$.
- ▶ The target posterior is a marginal of the PMMH stationary distribution.

Recall: tempering

- ▶ Problem: want to sample from $\pi(d\theta, dz)$, but this is hard.
- ▶ Idea: construct 'smooth' sequence $\{\pi_k(d\theta_k, dz_k)\}_{k=0}^K$ with π_0 easy to sample from and $\pi_K \approx \pi$.
- ▶ Sample from π_0 and 'propagate' samples across sequence to π_K .

Recall: tempering

- ▶ Problem: want to sample from $\pi(d\theta, dz)$, but this is hard.
- ▶ Idea: construct 'smooth' sequence $\{\pi_k(d\theta_k, dz_k)\}_{k=0}^K$ with π_0 easy to sample from and $\pi_K \approx \pi$.
- ▶ Sample from π_0 and 'propagate' samples across sequence to π_K .

Tempering scheme used

- ▶ Construct a decreasing sequence $(\lambda_k)_{k=0}^K$
- ▶ If the base state-space has observation law

$$y_t | x_t \sim N(g^\theta(x_t), \sigma^2),$$

let π_k be the posterior in the model with altered observation law

$$y_t | x_t \sim N(g^\theta(x_t), \sigma^2 + \lambda_k),$$

- ▶ This 'smoothes out' the observation densities

Recall: tempering

- ▶ Problem: want to sample from $\pi(d\theta, dz)$, but this is hard.
- ▶ Idea: construct 'smooth' sequence $\{\pi_k(d\theta_k, dz_k)\}_{k=0}^K$ with π_0 easy to sample from and $\pi_K \approx \pi$.
- ▶ Sample from π_0 and 'propagate' samples across sequence to π_K .

Tempering scheme used

- ▶ Construct a decreasing sequence $(\lambda_k)_{k=0}^K$
- ▶ If the base state-space has observation law

$$y_t|x_t \sim N(g^\theta(x_t), \sigma^2),$$

let π_k be the posterior in the model with altered observation law

$$y_t|x_t \sim N(g^\theta(x_t), \sigma^2 + \lambda_k),$$

- ▶ This 'smoothes out' the observation densities

Example

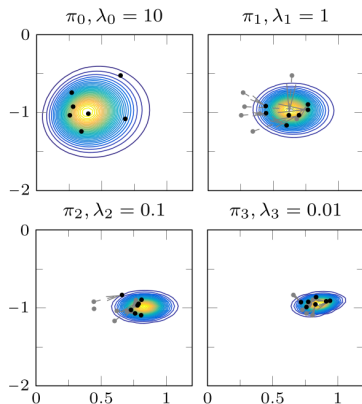
- ▶ State-space model with observation law

$$y_t|x_t \sim \delta_{g^\theta(x_t)}(dy_t)$$

- ▶ Likelihood and posterior intractable
- ▶ Temper with $\lambda_0 = 10$:

$$y_t|x_t, \lambda_0 \sim N(g^\theta(x_t), 10)$$

- ▶ Observations no longer highly informative, can easily sample using PMMH.



Example

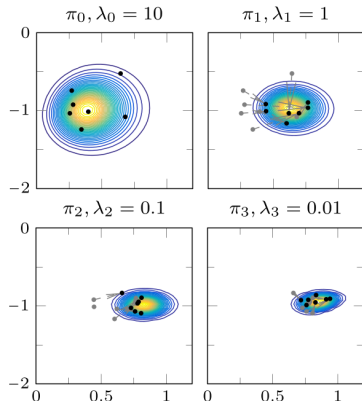
- ▶ State-space model with observation law

$$y_t | x_t \sim \delta_{g^\theta(x_t)}(dy_t)$$

- ▶ Likelihood and posterior intractable
- ▶ Temper with $\lambda_0 = 10$:

$$y_t | x_t, \lambda_0 \sim N(g^\theta(x_t), 10)$$

- ▶ Observations no longer highly informative, can easily sample using PMMH.



Example

- ▶ State-space model with observation law

$$y_t | x_t \sim \delta_{g^\theta(x_t)}(dy_t)$$

- ▶ Likelihood and posterior intractable
- ▶ Temper with $\lambda_0 = 10$:

$$y_t | x_t, \lambda_0 \sim N(g^\theta(x_t), 10)$$

- ▶ Observations no longer highly informative, can easily sample using PMMH.

Takeaway

Tempering can solve the weight degeneracy problem.

SMC construction

► Have: sequence of distributions $\pi_k(d\theta_k, dz_k)$ of the form

$$\pi_k(d\theta_k, dz_k) = \nu(d\theta) \frac{\gamma_k(\theta_k, dz_k)}{L_k}$$

with

$$\gamma_k(\theta_k, dz_k) = L_T^N(\theta_k, z_k | \lambda_k) K^{\theta_k}(dz_k | \lambda_k)$$

SMC construction

- Have: sequence of distributions $\pi_k(d\theta_k, dz_k)$ of the form

$$\pi_k(d\theta_k, dz_k) = \nu(d\theta) \frac{\gamma_k(\theta_k, dz_k)}{L_k}$$

with

$$\gamma_k(\theta_k, dz_k) = L_T^N(\theta_k, z_k | \lambda_k) K^{\theta_k}(dz_k | \lambda_k)$$

- Also have: PMMH kernels $M_k(d\theta_{k+1}, dz_{k+1}, \theta_k, z_k | \lambda_k)$ that leave invariant $\pi_k(d\theta_k, dz_k)$.

SMC construction

- Have: sequence of distributions $\pi_k(d\theta_k, dz_k)$ of the form

$$\pi_k(d\theta_k, dz_k) = \nu(d\theta) \frac{\gamma_k(\theta_k, dz_k)}{L_k}$$

with

$$\gamma_k(\theta_k, dz_k) = L_T^N(\theta_k, z_k | \lambda_k) K^{\theta_k}(dz_k | \lambda_k)$$

- Also have: PMMH kernels $M_k(d\theta_{k+1}, dz_{k+1}, \theta_k, z_k | \lambda_k)$ that leave invariant $\pi_k(d\theta_k, dz_k)$.
- So we can: construct a Feynman-Kac model with initial distribution

$$\mathbb{M}_0(d\theta_0, dz_0) = \pi_0(d\theta_0, dz_0),$$

transition kernels M_k , and potentials

$$G_k(d\theta_k, dz_k) = \frac{L_T^N(\theta_k, z_k | \lambda_k) K^{\theta_k}(dz_k | \lambda_k)}{L_T^N(\theta_k, z_k | \lambda_{k-1}) K^{\theta_k}(dz_k | \lambda_{k-1})}$$

SMC construction

- Have: sequence of distributions $\pi_k(d\theta_k, dz_k)$ of the form

$$\pi_k(d\theta_k, dz_k) = \nu(d\theta) \frac{\gamma_k(\theta_k, dz_k)}{L_k}$$

with

$$\gamma_k(\theta_k, dz_k) = L_T^N(\theta_k, z_k | \lambda_k) K^{\theta_k}(dz_k | \lambda_k)$$

- Also have: PMMH kernels $M_k(d\theta_{k+1}, dz_{k+1}, \theta_k, z_k | \lambda_k)$ that leave invariant $\pi_k(d\theta_k, dz_k)$.
- So we can: construct a Feynman-Kac model with initial distribution

$$\mathbb{M}_0(d\theta_0, dz_0) = \pi_0(d\theta_0, dz_0),$$

transition kernels M_k , and potentials

$$G_k(d\theta_k, dz_k) = \frac{L_T^N(\theta_k, z_k | \lambda_k) K^{\theta_k}(dz_k | \lambda_k)}{L_T^N(\theta_k, z_k | \lambda_{k-1}) K^{\theta_k}(dz_k | \lambda_{k-1})}$$

SMC construction: continued

- Can be shown that the marginals for this Feynman-Kac model are precisely the posteriors π_k .

SMC construction

- ▶ Have: sequence of distributions $\pi_k(d\theta_k, dz_k)$ of the form

$$\pi_k(d\theta_k, dz_k) = \nu(d\theta) \frac{\gamma_k(\theta_k, dz_k)}{L_k}$$

with

$$\gamma_k(\theta_k, dz_k) = L_T^N(\theta_k, z_k | \lambda_k) K^{\theta_k}(dz_k | \lambda_k)$$

- ▶ Also have: PMMH kernels $M_k(d\theta_{k+1}, dz_{k+1}, \theta_k, z_k | \lambda_k)$ that leave invariant $\pi_k(d\theta_k, dz_k)$.
- ▶ So we can: construct a Feynman-Kac model with initial distribution

$$\mathbb{M}_0(d\theta_0, dz_0) = \pi_0(d\theta_0, dz_0),$$

transition kernels M_k , and potentials

$$G_k(d\theta_k, dz_k) = \frac{L_T^N(\theta_k, z_k | \lambda_k) K^{\theta_k}(dz_k | \lambda_k)}{L_T^N(\theta_k, z_k | \lambda_{k-1}) K^{\theta_k}(dz_k | \lambda_{k-1})}$$

SMC construction: continued

- ▶ Can be shown that the marginals for this Feynman-Kac model are precisely the posteriors π_k .
- ▶ The corresponding forward-recursion is precisely the desired 'propagation of samples', can be implemented as a particle filter.

SMC construction

- ▶ Have: sequence of distributions $\pi_k(d\theta_k, dz_k)$ of the form

$$\pi_k(d\theta_k, dz_k) = \nu(d\theta) \frac{\gamma_k(\theta_k, dz_k)}{L_k}$$

with

$$\gamma_k(\theta_k, dz_k) = L_T^N(\theta_k, z_k | \lambda_k) K^{\theta_k}(dz_k | \lambda_k)$$

- ▶ Also have: PMMH kernels $M_k(d\theta_{k+1}, dz_{k+1}, \theta_k, z_k | \lambda_k)$ that leave invariant $\pi_k(d\theta_k, dz_k)$.

- ▶ So we can: construct a Feynman-Kac model with initial distribution

$$\mathbb{M}_0(d\theta_0, dz_0) = \pi_0(d\theta_0, dz_0),$$

transition kernels M_k , and potentials

$$G_k(d\theta_k, dz_k) = \frac{L_T^N(\theta_k, z_k | \lambda_k) K^{\theta_k}(dz_k | \lambda_k)}{L_T^N(\theta_k, z_k | \lambda_{k-1}) K^{\theta_k}(dz_k | \lambda_{k-1})}$$

SMC construction: continued

- ▶ Can be shown that the marginals for this Feynman-Kac model are precisely the posteriors π_k .
- ▶ The corresponding forward-recursion is precisely the desired 'propagation of samples', can be implemented as a particle filter.
- ▶ Theoretical properties (central limit theorems, stability) follow directly from Feynman-Kac model theory.

SMC construction

- ▶ Have: sequence of distributions $\pi_k(d\theta_k, dz_k)$ of the form

$$\pi_k(d\theta_k, dz_k) = \nu(d\theta) \frac{\gamma_k(\theta_k, dz_k)}{L_k}$$

with

$$\gamma_k(\theta_k, dz_k) = L_T^N(\theta_k, z_k | \lambda_k) K^{\theta_k}(dz_k | \lambda_k)$$

- ▶ Also have: PMMH kernels $M_k(d\theta_{k+1}, dz_{k+1}, \theta_k, z_k | \lambda_k)$ that leave invariant $\pi_k(d\theta_k, dz_k)$.
- ▶ So we can: construct a Feynman-Kac model with initial distribution

$$\mathbb{M}_0(d\theta_0, dz_0) = \pi_0(d\theta_0, dz_0),$$

transition kernels M_k , and potentials

$$G_k(d\theta_k, dz_k) = \frac{L_T^N(\theta_k, z_k | \lambda_k) K^{\theta_k}(dz_k | \lambda_k)}{L_T^N(\theta_k, z_k | \lambda_{k-1}) K^{\theta_k}(dz_k | \lambda_{k-1})}$$

SMC construction: continued

- ▶ Can be shown that the marginals for this Feynman-Kac model are precisely the posteriors π_k .
- ▶ The corresponding forward-recursion is precisely the desired 'propagation of samples', can be implemented as a particle filter.
- ▶ Theoretical properties (central limit theorems, stability) follow directly from Feynman-Kac model theory.

Takeaway

The proposed algorithm can be implemented as a particle filter on the extended parameter space $\Theta \times \mathcal{Z}$.

Adaptive tempering

► Want smooth tempering sequence; adapt temperature on the fly.

► Given $\lambda_{k-1}, \theta_{k-1}^n, z_{k-1}^n$, pick λ_k such that the weights W_k^n will give

$$\text{ESS} \approx \alpha \cdot N_\theta$$

for some fixed $\alpha \in [0, 1]$.

► Equation can be solved using a bisection method.

Algorithm: algorithm from Svensson et al.

Input: N_θ samples, N_x particles, initial temperature λ_0 , observations $y_{0:T}$

Output: approximate samples $\{(\theta^n, z^n)\}_{n=1}^{N_\theta}$ from $\pi(d\theta, dz)$.

```

1:  $k \leftarrow -1$ 
2: while  $\lambda_k > \lambda_{\min}$  do
3:    $k \leftarrow k + 1$ 
4:   if  $k = 0$  then
5:      $(\theta_0^n, z_0^n) \sim \pi_0(d\theta_0, dz_0)$ 
6:   else
7:      $A_k^{1:N_x} \sim \text{resample}(W_{k-1}^{1:N_x})$ 
8:      $(\theta_k^n, z_k^n) \sim M_k(\theta_{k-1}^{A_k^n}, z_{k-1}^{A_k^n}, d\theta_k, dz_k)$ 
9:   end if
10:   $\lambda_{k+1} \leftarrow \text{nxt\_temper}(\lambda_k, \theta_k^n, z_k^n)$ 
11:   $w_k^n \leftarrow \frac{L_T^{N_x}(\theta_k, z_k | \lambda_{k+1}) K^{\theta_k}(dz_k | \lambda_{k+1})}{L_T^{N_x}(\theta_k, z_k | \lambda_k) K^{\theta_k}(dz_k | \lambda_k)}$ 
12:   $W_k^n \leftarrow w_k^n / \sum_{m=1}^{N_\theta} w_k^m$ 

```

Adaptive tempering

- ▶ Want smooth tempering sequence; adapt temperature on the fly.
- ▶ Given $\lambda_{k-1}, \theta_{k-1}^n, z_{k-1}^n$, pick λ_k such that the weights W_k^n will give

$$\text{ESS} \approx \alpha \cdot N_\theta$$

for some fixed $\alpha \in [0, 1]$.

- ▶ Equation can be solved using a bisection method.

Computational note

- ▶ Note: if particle filter is bootstrap, $K^{\theta_k}(\text{d}z_k | \lambda_k)$ has density

$$k^\theta(z | \lambda_k) \propto \prod_{t=1}^{T-1} \prod_{n=1}^{N_x} \frac{f^\theta(y_t | X_t^{A_{t+1}^n}, \lambda_k)}{\sum_{m=1}^{N_x} f^\theta(y_t | X_t^n, \lambda_k)}$$

Algorithm: algorithm from Svensson et al.

Input: N_θ samples, N_x particles, initial temperature λ_0 , observations $y_{0:T}$

Output: approximate samples $\{(\theta^n, z^n)\}_{n=1}^{N_\theta}$ from $\pi(\text{d}\theta, \text{d}z)$.

- 1: $k \leftarrow -1$
- 2: **while** $\lambda_k > \lambda_{\min}$ **do**
- 3: $k \leftarrow k + 1$
- 4: **if** $k = 0$ **then**
- 5: $(\theta_0^n, z_0^n) \sim \pi_0(\text{d}\theta_0, \text{d}z_0)$
- 6: **else**
- 7: $A_k^{1:N_x} \sim \text{resample}(W_{k-1}^{1:N_x})$
- 8: $(\theta_k^n, z_k^n) \sim M_k(\theta_{k-1}^{A_k^n}, z_{k-1}^{A_k^n}, \text{d}\theta_k, \text{d}z_k)$
- 9: **end if**
- 10: $\lambda_{k+1} \leftarrow \text{nxt_temper}(\lambda_k, \theta_k^n, z_k^n)$
- 11: $w_k^n \leftarrow \frac{L_T^{N_x}(\theta_k, z_k | \lambda_{k+1}) K^{\theta_k}(\text{d}z_k | \lambda_{k+1})}{L_T^{N_x}(\theta_k, z_k | \lambda_k) K^{\theta_k}(\text{d}z_k | \lambda_k)}$
- 12: $W_k^n \leftarrow w_k^n / \sum_{m=1}^{N_\theta} w_k^m$

Challenging non-linear model

State-space model with parameter $\theta = (\theta_1, \theta_2)$, dynamics

$$x_0 = v_t,$$

$$x_t | x_{t-1} = \arctan(x_{t-1}) + \theta_1 u_t + v_t$$

$$y_t | x_t = |x_t| + \theta_1 \theta_2 + e_t$$

with $v_t \sim \mathcal{N}(0, 1)$, $e_t \sim \mathcal{N}(0.01)$, and u_t is some fixed vector (exogenous input).

Challenging non-linear model

State-space model with parameter $\theta = (\theta_1, \theta_2)$, dynamics

$$x_0 = v_t,$$

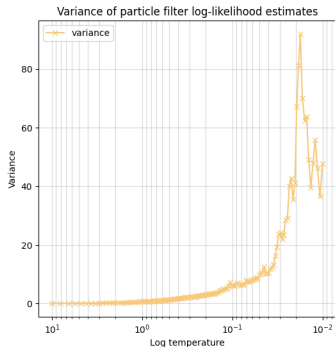
$$x_t | x_{t-1} = \arctan(x_{t-1}) + \theta_1 u_t + v_t$$

$$y_t | x_t = |x_t| + \theta_1 \theta_2 + e_t$$

with $v_t \sim N(0, 1)$, $e_t \sim N(0.01)$, and u_t is some fixed vector (exogenous input).

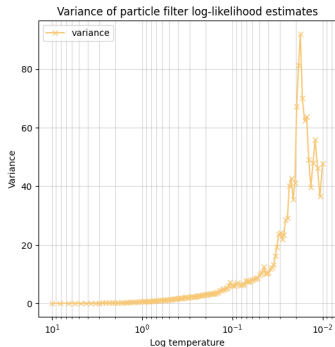
Little experiment

Set true parameter $\theta_0 = (1, 1)$, generate observations $y_{0:200}$, run SMC with $N_x = 100$, $N_\theta = 100$, starting temperature $\lambda_0 = 10$, uniform prior on $[0, 2] \times [0, 2]$, bootstrap particle filter.



Little experiment

Set true parameter $\theta_0 = (1, 1)$, generate observations $y_{0:200}$, run SMC with $N_x = 100, N_\theta = 100$, starting temperature $\lambda_0 = 10$, uniform prior on $[0, 2] \times [0, 2]$, bootstrap particle filter.



Little experiment

Set true parameter $\theta_0 = (1, 1)$, generate observations $y_{0:200}$, run SMC with $N_x = 100, N_\theta = 100$, starting temperature $\lambda_0 = 10$, uniform prior on $[0, 2] \times [0, 2]$, bootstrap particle filter.

Takeaway

- Variance stabilized for most of the run, weight degeneracy is avoided.
- Variance still unstable in low temperature regime. Underscores importance of early stopping

Outlook

- ▶ Combine with other variance reduction techniques in the highly informative setting, like the bridged particle filter (Del Moral and Murray)
- ▶ Generalize to state-space models with arbitrary a.c. observation kernel $F_t^\theta(dy_t, x_t)$ (could generalize by tempering via convolution of the observation kernel with a Gaussian $N(0, \lambda_k)$)
- ▶ Try alternative criteria for adaptive tempering, rather than just ESS
- ▶ More robust numerical evaluation.

The End

Thanks for listening

Any questions?